

Introduction to GitHub

Module 1: Introduction to Git & GitHub

1. What is Git?

- Distributed Version Control System
- Tracks changes in source code
- Helps in collaboration

2. What is GitHub?

- Cloud-based hosting service for Git repositories
 - Allows collaboration, sharing, and teamwork
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Module 2: Working with Remotes

1. View Existing Remotes

```
git remote -v
```

- Shows the URLs of remote repositories connected to your local repo.

2. Add a Remote Repository

```
git remote add origin https://github.com/username/repository.git  
git remote -v
```

- *origin* is the default alias name for the remote repository.
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Module 3: Branching & Pushing Code

1. Rename the Default Branch

```
git branch -M main
```

- Renames your current branch to *main*.

2. Push Code for the First Time

```
git push -u origin main
```

- *-u* sets *origin main* as the default upstream.
- After this, just use:

```
git push
```

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3. Push to Other Branches

```
git checkout dev
git push -u origin dev

git checkout mukesh/multiply
git push -u origin mukesh/multiply
```

- Each branch must be pushed separately to GitHub.
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Module 4: Workflow Example

1. Make a Change

```
git add Diff.java
git commit -m "Added implementation for Diff.java"
git push
```

- Commits local changes and pushes them to GitHub.
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Module 5: Collaboration on GitHub

1. Adding Collaborators

- Go to **GitHub Repository** → **Settings** → **Manage Access**
- Click **Invite Collaborator**
- Enter their GitHub username or email ID.
- They can now **clone**, **modify**, and **push**.

2. Fork & Pull Request Workflow

- **Fork**: Copy someone else's repository to your account.
- **Clone**: Bring it to your local system.
- **Make Changes & Commit**
- **Push changes to your fork**
- **Create Pull Request** to the original repository.